

Energy Transfer

Test · 25 marks total

Name: _____

Date: _____

Score: _____ / 25

Section A — Key ideas

A1. Name the four ways that energy can be transferred.

[4 marks]

A2. State the principle of conservation of energy.

[1 mark]

Section B — Kinetic & gravitational potential energy

B1. Calculate the kinetic energy of a 1200 kg car travelling at 15 m/s.

[2 marks]

B2. A ball of mass 0.4 kg is lifted 3 m. Calculate its gravitational potential energy. ($g = 9.8 \text{ N/kg}$)

[2 marks]

Section C — Elastic potential energy & specific heat capacity

C1. A spring with a spring constant of 25 N/m is stretched by 0.08 m. Calculate its elastic potential energy. Show your working.

[3 marks]

C2. 2 kg of water gains 42,000 J of energy when heated. The specific heat capacity of water is 4200 J/kg°C. Calculate the temperature rise. Show how you rearranged the equation.

[3 marks]

Section D — Power & efficiency

D1. A kettle has a power rating of 2000 W. Calculate the energy it transfers when used for 3 minutes.

[3 marks]

D2. A lamp has a total energy input of 60 J, and produces 12 J of useful light energy. Calculate its efficiency as a percentage.

[2 marks]

Section E — Extended reasoning

E1. Explain, in terms of energy, why no real machine can ever be 100% efficient, and give one way of reducing unwanted energy transfers caused by friction.

[3 marks]

Section F — Energy stores

State the main energy store involved in each of the following:

(a) a stretched elastic band, before release (b) petrol in a car's fuel tank

[2 marks]